

PLANT DISEASE DETECTION USING CONVOLUTIONAL NEURAL NETWORKS

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Year: 2024

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Degree Program: BSCS

Keywords: CNN, plant disease, deep learning, image classification

ABSTRACT

This study presents a CNN-based system for automated plant disease detection from leaf images. Using ResNet-50 on a dataset of 5,000 annotated images, the system achieved 93.4% accuracy, outperforming SVM-based baselines by 6.4%.

INTRODUCTION

Agriculture is a critical sector of the Philippine economy, contributing nearly 10% to GDP. Early detection of leaf diseases such as Bacterial Blight, Powdery Mildew, and Late Blight can prevent up to 25% yield losses annually. Manual inspection by agronomists is slow and costly, motivating automation. This study aims to: (1) develop a CNN model for multi-class leaf disease classification; (2) evaluate its performance on a curated local dataset; (3) compare results with traditional machine learning classifiers.

METHODOLOGY

This study employed a ResNet-50 Convolutional Neural Network pretrained on ImageNet and fine-tuned on a custom dataset of 5,000 leaf images from the PlantVillage corpus. The dataset covers 10 disease categories. An 80/20 stratified split was used for training and validation. Data augmentation included horizontal flip, 15-degree rotation, and brightness jitter. The model was trained for 50 epochs using Adam optimizer ($\text{lr}=0.001$) with early stopping ($\text{patience}=5$). A confusion matrix, precision, recall, and F1-score were computed for evaluation. GPU: NVIDIA GTX 1060.

RESULTS

The ResNet-50 model achieved 93.4% overall accuracy on the validation set. Precision was 92.1%, recall 93.8%, and F1-score 92.9%. Per-class accuracy ranged from 87% (Early Blight vs Late Blight confusion) to 97% (Powdery Mildew). Training converged at epoch 38. Comparison with SVM (87%) and KNN (79%) classifiers confirmed the superiority of the CNN approach.

DISCUSSION

The 93.4% accuracy confirms CNNs as the most effective approach for automated plant disease detection. The main source of error is visual similarity between Early Blight and Late

Blight, a known challenge in the domain. Data augmentation significantly improved generalization. Future work should explore Vision Transformers and domain-specific pretraining using Philippine crop datasets.